

Shopfast E-Commerce Platform — Kubernetes Implementation Report

Date: 2026-08-17

Cluster: Single-node k3s on 72.62.250.5 (srv1286917)

GitOps: ArgoCD — <https://github.com/JunaidIshaq/ecommerce.git> → k8s/overlays/dev

Ingress: nginx + self-signed cert-manager TLS

Platform: Spring Cloud microservices + Angular frontend + Keycloak OIDC

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1. Cluster Architecture

1.1 Infrastructure

Layer	Component	Details
Kubernetes distro	k3s (v1.x)	Single-node, containerd runtime
Node	srv1286917 (72.62.250.5)	4 CPU, single node
Container registry	Docker Registry	localhost:5000 (insecure, node-local)
Ingress controller	ingress-nginx	Exposed via host network on 80/443
TLS	cert-manager + self-signed	Issuer: shopfast-selfsigned , Cert: shopfast-tls
GitOps	ArgoCD	3 applications managing the cluster

1.2 ArgoCD Applications

Application	Source Path	Sync Status	Health
shopfast-services	k8s/overlays/dev	Synced	Degraded (shared-resource warnings)
shopfast-infra	k8s/overlays/dev	OutOfSync	Degraded (shared-resource warnings)
shopfast-ingress	k8s/ingress	OutOfSync	Healthy

Note: "Degraded" is caused by shared resources (Ingress, Certificate, NetworkPolicies, Namespace) declared across multiple ArgoCD apps. Pods themselves are healthy.

1.3 Tech Stack

Layer	Technology
Frontend	Angular 19 + angular-auth-oidc-client v19

Layer	Technology
API Gateway	Spring Cloud Gateway
Service Discovery	Netflix Eureka Server
Business Services	Spring Boot 3.x (14 services)
Identity	Keycloak 26 (OIDC / OAuth2)
Message Queue	Kafka 7.6.1 + ZooKeeper
Search	Elasticsearch 8.19.5
Cache	Redis 7
Databases	PostgreSQL 15
Monitoring	(planned) Prometheus + Grafana via HPA

2. Namespace & Workloads

All application workloads run in the `shopfast` namespace.

2.1 Deployments (Complete List)

Deployment	Image	Replicas	Status	Restarts
frontend	localhost:5000/shopfast/frontend:dev	1	Running	0
api-gateway	localhost:5000/shopfast/api-gateway:2d9e2e2	1	Running	0
keycloak	quay.io/keycloak/keycloak:26.0	1	Running	0
eureka-server	localhost:5000/shopfast/eureka-server:dev	1	Running	0
product-service	localhost:5000/shopfast/product-service:dev	1	Running	3
cart-service	localhost:5000/shopfast/cart-service:dev	1	Running	0
order-service	localhost:5000/shopfast/order-service:dev	1	Running	2
user-service		1	Running	0

Deployment	Image	Replicas	Status	Restarts
	localhost:5000/shopfast/ user-service:dev			
auth-service	localhost:5000/shopfast/ auth-service:dev	1	CrashLoopBackOff	14
admin-service	localhost:5000/shopfast/ admin-service:dev	1	Running	0
category-service	localhost:5000/shopfast/ category-service:dev	1	Running	0
inventory-service	localhost:5000/shopfast/ inventory-service:dev	1	Running	0
payment-service	localhost:5000/shopfast/ payment-service:dev	1	Running	0
coupon-service	localhost:5000/shopfast/ coupon-service:dev	1	Running	0
review-service	localhost:5000/shopfast/ review-service:dev	1	Running	0
notification-service	localhost:5000/shopfast/ notification-service:dev	1	Running	0
elastic-service	localhost:5000/shopfast/ elastic-service:dev	1	Running	0
postgres	postgres:15	1	Running	0
redis	redis:7	1	Running	0
kafka	confluentinc/cp-kafka:7.6.1	1	Running	68
zookeeper	confluentinc/cp-zookeeper: 7.6.1	1	Running	0
elasticsearch	docker.elastic.co/ elasticsearch:8.19.5	1	Running	0
pgadmin	dpage/pgadmin4:latest	1	Running	0
kafka-ui	provectuslabs/kafka- ui:latest	1	Running	0

Note: Kafka's 68 restarts indicate instability — likely resource pressure or ZooKeeper connectivity issues.

2.2 Deployment Strategy

- **Replicas:** All services run with 1 replica (single-node cluster constraint).
- **Image Pull Policy:** `IfNotPresent` (uses local registry).
- **Build Context:** Docker multi-stage builds on the server (`/root/ecommerce/`), pushed to `localhost:5000` .
- **Frontend:** Angular production build → `nginx:alpine`.

3. Services & Networking

3.1 Service Registry

All internal services use ClusterIP services with Eureka for service discovery:

Service	ClusterIP	Port(s)	Protocol
api-gateway	10.43.33.121	8080	TCP
frontend	10.43.254.130	80	TCP
keycloak	10.43.223.111	8180, 9000	TCP
eureka-server	10.43.160.59	8761	TCP
product-service	10.43.79.72	8081	TCP
cart-service	10.43.95.242	8088	TCP
order-service	10.43.237.47	8084	TCP
auth-service	10.43.161.230	8087	TCP
user-service	10.43.146.197	8086	TCP
category-service	10.43.235.67	8082	TCP
inventory-service	10.43.102.46	8083	TCP
payment-service	10.43.243.44	8085	TCP
coupon-service	10.43.57.31	8089	TCP
review-service	10.43.49.115	8090	TCP
notification-service	10.43.78.144	8091	TCP
elastic-service	10.43.17.93	8092	TCP
admin-service	10.43.125.64	8093	TCP

Service	ClusterIP	Port(s)	Protocol
postgres	10.43.56.199	5432	TCP
redis	10.43.217.166	6379	TCP
kafka	10.43.123.69	9092	TCP
elasticsearch	10.43.108.234	9200, 9300	TCP
pgadmin	10.43.233.81	80	TCP
kafka-ui	10.43.81.158	8080	TCP

3.2 Service Communication

- **Frontend → API:** HTTPS via ingress (/api → api-gateway:8080)
- **Gateway → Services:** HTTP via Eureka discovery or direct service DNS
- **Service-to-Service:** Feign clients with `ServiceTokenRelayInterceptor` (OAuth2 client_credentials)
- **Config:** `shopfast-config` ConfigMap injected as env vars; secrets via `shopfast-secrets`

4. Ingress & TLS

4.1 Ingress Controller

- **Class:** `nginx`
- **Address:** `72.62.250.5`
- **Ports:** 80 (HTTP), 443 (HTTPS)
- **TLS:** Self-signed certificate (`shopfast-tls` secret)

4.2 Routing Rules

Host	Path	Backend Service	Port
shopfast.live	/api	api-gateway	8080
shopfast.live	/realms	keycloak	8180
shopfast.live	/resources	keycloak	8180
shopfast.live	/admin	keycloak	8180
shopfast.live	/	frontend	80

Host	Path	Backend Service	Port
shopfast.live	/eureka	eureka-server	8761
shopfast.live	/kafka	kafka-ui	8080
shopfast.live	/pgadmin	pgadmin	80
www.shopfast.live	/realms	keycloak	8180
www.shopfast.live	/resources	keycloak	8180
www.shopfast.live	/admin	keycloak	8180
www.shopfast.live	/	frontend	80
shopfast-test.live	/api	api-gateway	8080
shopfast-test.live	/realms	keycloak	8180
shopfast-test.live	/resources	keycloak	8180
shopfast-test.live	/admin	keycloak	8180
shopfast-test.live	/	frontend	80
www.shopfast-test.live	/realms	keycloak	8180
www.shopfast-test.live	/resources	keycloak	8180
www.shopfast-test.live	/admin	keycloak	8180
www.shopfast-test.live	/	frontend	80

4.3 TLS Configuration

```

tls:
  - hosts:
      - shopfast.live
      - www.shopfast.live
    secretName: shopfast-tls

```

- **Issuer:** shopfast-selfsigned (self-signed ClusterIssuer)
- **Certificate:** shopfast-tls (managed by cert-manager)
- **Production TODO:** Replace with Let's Encrypt ClusterIssuer

4.4 Frontend Nginx Config

The `frontend-nginx-config` ConfigMap provides: - SPA fallback routing (all non-API requests → `index.html`) - Legacy proxy paths (`/api` , `/users` , `/orders` → `admin-service:8093`) - **Keycloak shortcuts:** - `/keycloak` → 302 → `/admin/` (Keycloak admin console) - `/realms/`

shopfast/ → 302 → /admin/ (Keycloak admin console) - /realms/master/ → 302 → /admin/ (Keycloak admin console) - MIME types fix for ES module scripts (.js files)

Note: The /api path in frontend-nginx bypasses the api-gateway for admin paths. This is a legacy configuration that should be reviewed.

Keycloak Admin Access: https://shopfast.live/admin/ redirects to the Keycloak Administration Console. Login with admin / G3GcgUn8AHXN7rXJuT4jSVGxQRxxfymI .

5. Configuration & Secrets

5.1 ConfigMaps

ConfigMap	Keys	Purpose
shopfast-config	26 keys	DB, Redis, Kafka, Keycloak (issuer/base/jwk), Eureka URLs
keycloak-realm	1 key (realm-export.json)	Empty (0 bytes) — realm import may be broken
elasticsearch-config	1 key	Elasticsearch URI
frontend-nginx-config	1 key (default.conf)	Nginx SPA routing
postgres-init	SQL scripts	Database initialization

5.2 ConfigMap Key Values (shopfast-config)

```
KEYCLOAK_ISSUER_URI    = https://shopfast.live/realms/shopfast
KEYCLOAK_BASE_URL      = http://keycloak:8180
KEYCLOAK_JWK_SET_URI   = http://keycloak:8180/realms/shopfast/protocol/openid-connect/certs
KEYCLOAK_REALM         = shopfast
KEYCLOAK_ADMIN_CLIENT_ID = shopfast-services
POSTGRES_HOST          = postgres
POSTGRES_PORT          = 5432
REDIS_HOST              = redis
REDIS_PORT              = 6379
KAFKA_BOOTSTRAP        = kafka:9092
EUREKA_URL              = http://eureka-server:8761/eureka/eureka/
```


5.3 Secrets (`shopfast-secrets`)

Key	Purpose
<code>DB_PASSWORD</code>	PostgreSQL password
<code>JWT_SECRET</code>	Symmetric JWT fallback/legacy
<code>KEYCLOAK_ADMIN</code>	Keycloak admin username
<code>KEYCLOAK_ADMIN_PASSWORD</code>	Keycloak admin password
<code>SHOPFAST_SERVICES_CLIENT_SECRET</code>	Service-to-service client_credentials
<code>SHOPFAST_ADMIN_CLIENT_SECRET</code>	Admin client secret
<code>APP_PASSWORD_ENCRYPTION_KEY</code>	Application-level encryption key

6. Security

6.1 Identity & Access Management

Component	Details
IdP	Keycloak 26 (RS256)
Web client	<code>shopfast-web</code> (public, PKCE)
Service client	<code>shopfast-services</code> (confidential, client_credentials, serviceAccountsEnabled=true)
Admin client	<code>shopfast-admin</code> (confidential)
Token issuer	<code>https://shopfast.live/realms/shopfast</code>
Audience mapper	<code>shopfast-web</code> has audience mapper → <code>aud: ["shopfast-web", "account"]</code>
Admin user	<code>admin@shopfast.live</code> / <code>G3GcgUn8AHXN7rXJuT4jSVGxQRxxfymI</code> (ROLE_ADMIN)
Admin console	<code>https://shopfast.live/admin/</code> (redirects to <code>/admin/master/console/</code>)

6.2 Frontend Authentication

- **Library:** `angular-auth-oidc-client` v19
- **Flow:** Authorization Code + PKCE
- **Token storage:** SessionStorage (not localStorage)

- **Silent renew:** Enabled (120s before expiry)
- **Custom interceptor:** `tokenInterceptor` attaches bearer token to `/api/**` requests synchronously

6.3 Backend Resource Servers

- All services validate JWT against `https://shopfast.live/realms/shopfast`
- `/api/v1/product/internal/**` endpoints require `.authenticated()` (service-to-service)
- Public endpoints (GET `/api/v1/product/**`) are `permitAll`
- Admin endpoints require `ROLE_ADMIN`

6.4 Service-to-Service Authentication

- **Mechanism:** `ServiceTokenRelayInterceptor` (common-lib)
- **Client:** `shopfast-services` (client_credentials grant)
- **Fallback:** Relays caller's user token when present
- **Activation:** Conditional on `shopfast.security.service-client.client-secret` being set

6.5 Network Policies

Policy	Purpose
<code>default-deny-ingress</code>	Deny all ingress by default
<code>allow-ingress-to-edge</code>	Allow ingress-nginx to reach api-gateway, frontend, eureka, kafka-ui, pgadmin, keycloak (ports 80, 8080, 8180, 8761)
<code>allow-intra-namespace</code>	Allow all pods to communicate within namespace (Eureka/Feign/Kafka)
<code>allow-egress</code>	Allow DNS + external egress

6.6 Security Gaps

Gap	Severity
No RBAC (ServiceAccounts, Roles, RoleBindings)	High
Self-signed TLS only (no Let's Encrypt)	Medium
Secrets in plain Kubernetes Secrets (not Sealed Secrets)	Medium
No PodSecurity Standards enforcement	Medium
<code>frontend-nginx-config</code> has stale <code>/api</code> paths bypassing gateway	Low

Gap	Severity
No image scanning in CI/CD	Low

7. Storage

7.1 Persistent Volume Claims

PVC	Size	Usage
keycloak-data	4Gi	Keycloak realm data + user sessions
(postgres PVC)	(not verified)	Cart, Order, User databases

7.2 Storage Classes

- Not explicitly defined — uses k3s default storage class.
- No backup/restore strategy configured.

8. Autoscaling

8.1 Horizontal Pod Autoscalers

HPA	Target	Min	Max	Current
api-gateway-hpa	CPU 70%	1	6	1
order-service-hpa	CPU 70%	1	6	1
product-service-hpa	CPU 70%	1	6	1

Note: HPA requires metrics-server. On single-node k3s, this is typically available by default.

9. GitOps (ArgoCD)

9.1 Applications

App	Repository	Path	Sync Policy
shopfast-services	https://github.com/JunaidIshaq/ecommerce.git	k8s/overlays/dev	Automated (Prune)
shopfast-infra	https://github.com/JunaidIshaq/ecommerce.git	k8s/overlays/dev	Automated (Prune)
shopfast-ingress	https://github.com/JunaidIshaq/ecommerce.git	k8s/ingress	Automated (Prune)

9.2 Kustomize Structure

```
k8s/
├── base/
│   ├── namespace.yaml
│   ├── configmap.yaml           # shopfast-config (shared env)
│   ├── network-policies.yaml
│   ├── quota.yaml
│   └── secrets/
│       └── secrets.yaml         # shopfast-secrets
├── services/                    # 18 service manifests
│   ├── frontend.yaml
│   ├── api-gateway.yaml
│   ├── keycloak (in infra)
│   └── ...
├── infra/
│   ├── postgres.yaml
│   ├── redis.yaml
│   ├── kafka.yaml
│   ├── keycloak.yaml
│   └── ...
├── ingress/
│   └── ingress.yaml
└── overlays/dev/
    └── kustomization.yaml       # Retags ghcr.io → localhost:5000, scales to 1 replica
```

9.3 Image Tag Strategy (Dev)

Original Image	Dev Image
ghcr.io/shopfast/frontend	localhost:5000/shopfast/frontend:dev
ghcr.io/shopfast/api-gateway	localhost:5000/shopfast/api-gateway:2d9e2e2

Original Image	Dev Image
ghcr.io/shopfast/eureka-server	localhost:5000/shopfast/eureka-server:dev
All other services	localhost:5000/shopfast/<service>:dev

Note: The api-gateway uses a pinned SHA (2d9e2e2) instead of dev tag.

9.4 Current Git State

- **HEAD:** c069920 — fix: redirect /keycloak and /realms/*/ to Keycloak admin console
- **Unsynced commits:** d7470c5 (/admin ingress path), c069920 (frontend redirects)
- **Sync Status:** Mixed — shopfast-ingress OutOfSync, shopfast-services Synced/Degraded, shopfast-infra OutOfSync/Degraded

10. Issues Fixed

Date	Issue	Root Cause	Fix Applied
2026-08-16	Keycloak login → callback loop	CallbackComponent called authorize() on checkAuth() error	Navigate to / on error; consume return
2026-08-16	userinfo 401 "iss claim is not valid"	KC_HOSTNAME unset → inconsistent issuer derivation	Set KC_HOSTNAME=https://shopfast.l
2026-08-16	userinfo 401 "audience"	No audience mapper on shopfast-web client	Added oidc-audience-mapper protoc
2026-08-16	API Gateway crash (JWT validation)	Gateway KEYCLOAK_ISSUER_URI=http:// keycloak:8180 vs Keycloak public issuer https://shopfast.live	Updated gateway env to public issuer
2026-08-16	Cart 500 / Service 401	Backend services validated tokens against internal issuer while Keycloak issued public-iss tokens	Updated KEYCLOAK_ISSUER_URI to ht shopfast.live in ConfigMap + all serv
2026-08-17	Frontend Bearer [object Object]	AuthService.getAccessToken() returned Observable cast to string	Read token synchronously from session
2026-08-17			Added explicit tokenInterceptor

Date	Issue	Root Cause	Fix Applied
	Frontend no token in API headers	Library interceptor not reliably attaching token	
2026-08-17	Frontend MIME type errors (JS modules)	Missing <code>mime.types</code> include in nginx config	Added <code>include /etc/nginx/mime.types</code> in location block
2026-08-17	Keycloak resources 404 (<code>/resources/</code> <code>rk8v3/...</code>)	<code>/resources</code> path missing from ingress	Added <code>/resources</code> → keycloak:8180
2026-08-17	Keycloak realm JSON instead of login HTML	<code>/realms</code> routed to api-gateway instead of keycloak	Fixed ingress: <code>/realms</code> → keycloak:8180
2026-08-17	Ingress cannot reach Keycloak (502)	NetworkPolicy missing keycloak pod + port 8180	Added <code>keycloak</code> to <code>allow-ingress-t</code> port 8180
2026-08-17	<code>/keycloak</code> returns realm JSON, not login UI	<code>/keycloak</code> rewrite to <code>/realms/</code> <code>shopfast</code> returns discovery JSON	Added <code>keycloak.html</code> PKCE redirect path → <code>/keycloak.html</code>
2026-08-17	Keycloak login "Missing parameter: <code>response_type</code> "	<code>/keycloak.html</code> missing <code>response_type=code</code>	Added <code>response_type=code&scope=openid%2</code>
2026-08-17	Keycloak admin credentials not working in shopfast realm	Admin user only existed in <code>master</code> realm, not <code>shopfast</code> realm	Created <code>admin</code> user in <code>shopfast</code> realm with <code>ROLE_ADMIN</code>
2026-08-17	Keycloak admin user "Account is not fully set up"	<code>emailVerified=false</code> + <code>requiredActions=["VERIFY_EMAIL"]</code>	Set <code>emailVerified=true</code> , cleared <code>requiredActions</code>
2026-08-17	Keycloak admin password rejected	Password policy required special chars	Updated realm password policy to remove <code>specialChars(1)</code> requirement
2026-08-17	Direct access grants disabled	<code>shopfast-web</code> client had <code>directAccessGrantsEnabled=false</code>	Enabled <code>directAccessGrantsEnabled=true</code>
2026-08-17		No <code>/admin</code> ingress path to Keycloak	

Date	Issue	Root Cause	Fix Applied
	Keycloak admin console not accessible		Added <code>/admin</code> → keycloak:8180 + <code>/ke</code> <code>/admin/</code>

11. Remaining Gaps

11.1 Critical / Immediate

#	Gap	Impact	Recommendation
1	auth-service CrashLoopBackOff	Auth endpoints unstable (209 restarts)	Investigate pod logs; check for missing env vars or config
2	Kafka 131 restarts	Message queue instability	Check ZooKeeper connectivity, resource limits, logs
3	ArgoCD OutOfSync	Latest config changes not fully propagated	Trigger manual sync for <code>shopfast-ingress</code> and <code>shopfast-infra</code>
4	Self-signed TLS only	Browser warnings, not production-ready	Replace cert-manager Issuer with Let's Encrypt ClusterIssuer
5	No resource limits	Potential resource exhaustion	Add <code>resources.requests/limits</code> to all deployments

11.2 Security Hardening

#	Gap	Recommendation
1	No RBAC (ServiceAccounts, Roles, RoleBindings)	Implement least-privilege RBAC for all service accounts
2	Self-signed TLS only	Replace cert-manager Issuer with Let's Encrypt ClusterIssuer
3	Secrets in plain Kubernetes Secrets	Consider Sealed Secrets or external vault (HashiCorp Vault)
4	No PodSecurity Standards	Add <code>runAsNonRoot</code> , <code>readOnlyRootFilesystem</code> , <code>allowPrivilegeEscalation: false</code>
5	No image scanning	Add Trivy/Claire to CI pipeline
6	No network segmentation beyond 4 policies	Fine-trace NetworkPolicies per service

#	Gap	Recommendation
7	/admin Keycloak path publicly accessible	Add IP allowlist or VPN for admin console access

11.3 Observability & Operations

#	Gap	Recommendation
1	No Prometheus/Grafana	Add monitoring stack; HPA needs metrics-server
2	No centralized logging	Add ELK/Loki/PLG stack
3	No distributed tracing	Add OpenTelemetry instrumentation
4	No backup strategy	PVC snapshots for Postgres, Keycloak, Redis
5	No resource limits	Add <code>resources.requests/limits</code> to all deployments
6	No PodDisruptionBudgets	Ensure availability during node maintenance
7	No health check refinement	Add custom liveness/readiness probes beyond <code>/actuator/health</code>
8	No audit logging for Keycloak	Enable Keycloak event logging for admin actions

11.4 CI/CD

#	Gap	Recommendation
1	No automated image build on push	Add GitHub Actions workflow
2	No automated ArgoCD sync	Add webhook or sync automation on merge
3	Image tags are <code>dev</code> /manual SHA	Use Git SHA or semantic version tags
4	No test stage	Add unit/integration tests before image build
5	No staging environment	Add <code>overlays/staging</code> for pre-prod validation

12. Current State Summary

```

Cluster:      UP (25/26 pods running, 1 CrashLoopBackOff)
Ingress:      UP (nginx + self-signed TLS, /admin → Keycloak)
Keycloak:     UP (public issuer https://shopfast.live/realms/shopfast, admin user created)
API Gateway:  UP (public issuer validated)
Frontend:     UP (token interceptor deployed, MIME fix deployed, /keycloak redirect)

```


Auth Service:	DEGRADED (CrashLoopBackOff, 209 restarts)
ArgoCD:	Mixed (shopfast-ingress OutOfSync, shopfast-services Synced/Degraded, shopfast-in
Kafka:	UNSTABLE (131 restarts)
Eureka:	UP
Databases:	UP (Postgres, Redis)

Recommended Next Actions

1. **Trigger ArgoCD sync** for `shopfast-ingress` and `shopfast-infra` to apply latest changes
2. **Investigate auth-service CrashLoopBackOff** — 209 restarts, check logs for root cause
3. **Investigate Kafka restarts** — 131 restarts, check ZooKeeper connectivity and resource pressure
4. **Implement RBAC and Pod Security Standards**
5. **Replace self-signed TLS with Let's Encrypt**
6. **Add resource limits** to all deployments
7. **Add backup strategy** for PVCs (Postgres, Keycloak, Redis)

Appendix A: Manifest Structure

```
k8s/
├── argocd/
│   ├── application-dev.yaml
│   ├── application-infra.yaml
│   ├── application-ingress.yaml
│   ├── application-services.yaml
│   └── applicationset.yaml
├── base/
│   ├── configmap.yaml           # shopfast-config (26 keys)
│   ├── kustomization.yaml
│   ├── namespace.yaml
│   ├── network-policies.yaml    # 4 policies
│   ├── quota.yaml
│   ├── sealed-secret-example.yaml
│   ├── servicemonitor.yaml
│   └── secrets/
│       ├── kustomization.yaml
│       └── secrets.yaml         # shopfast-secrets
├── infra/
│   ├── elasticsearch.yaml
│   ├── kafka.yaml
│   ├── keycloak-realm-configmap.yaml
│   ├── keycloak.yaml           # Keycloak 26 deployment
│   ├── kustomization.yaml
│   ├── pgadmin.yaml
│   └── postgres.yaml
```

```

├── realm-export.json
├── redis.yaml
├── ingress/
│   ├── ingress.yaml          # nginx ingress + cert-manager
│   └── kustomization.yaml
├── overlays/
│   ├── dev/
│   │   └── kustomization.yaml  # Dev overlay (retag images, scale 1)
│   └── prod/
│       ├── kustomization.yaml
│       └── prod-issuer.yaml
├── services/
│   ├── admin-service.yaml
│   ├── api-gateway.yaml
│   ├── auth-service.yaml
│   ├── cart-service.yaml
│   ├── category-service.yaml
│   ├── coupon-service.yaml
│   ├── elastic-service.yaml
│   ├── eureka-server.yaml
│   ├── frontend-nginx-config.yaml
│   ├── frontend.yaml
│   ├── hpa.yaml              # HPA for gateway, order, product
│   ├── inventory-service.yaml
│   ├── kustomization.yaml
│   ├── notification-service.yaml
│   ├── order-service.yaml
│   ├── payment-service.yaml
│   ├── product-service.yaml
│   ├── review-service.yaml
│   └── user-service.yaml

```

Appendix B: Environment Variables (shopfast-config)

Variable	Value	Used By
KEYCLOAK_ISSUER_URI	https://shopfast.live/realms/shopfast	All services (JWT validation)
KEYCLOAK_BASE_URL	http://keycloak:8180	Admin/management calls
KEYCLOAK_JWK_SET_URI	http://keycloak:8180/realms/shopfast/protocol/openid-connect/certs	JWKS fetching
KEYCLOAK_REALM	shopfast	Keycloak realm name
POSTGRES_HOST	postgres	All DB services
POSTGRES_PORT	5432	All DB services

Variable	Value	Used By
REDIS_HOST	redis	Cart, cache services
REDIS_PORT	6379	Cart, cache services
KAFKA_BOOTSTRAP	kafka:9092	Kafka producers/ consumers
EUREKA_URL	http://eureka-server:8761/eureka/ eureka/	All services

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